MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCNBC100 - BIOCHEMISTRY OF ENERGYTRANSFORMATION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation

Part A
MCQ
Answer all questions
Each question carries 1 mark

1. Uniport
2. Redox Potential
3. Fischers theory
4. Apoenzyme
5. Phosphoenolpyruvate
6. hnRNA (heterogeneous nuclear RNA)

Part B

Short Answer Questions

Answer any 2 questions

Each question carries 2 marks

7. **Define Osmosis.*(Any one point 2 mark)***

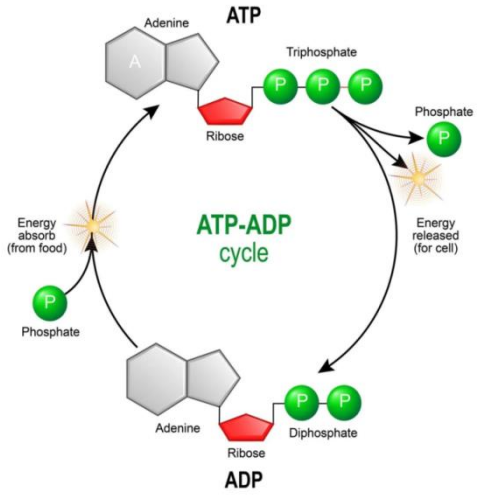
- Osmosis is the diffusion of water through a semi-permeable membraneaccording to its concentration gradient which is inversely proportional to the concentration of solutes.
- Water will move from an area with a higher concentration of water to the other side of the membrane with a lower concentration of water.

8. Describe ATP-ADP cycle.

The ATP–ADP cycle is the process by which energy is stored and released in cells. When energy is needed, ATP is broken down into ADP and inorganic phosphate (Pi), releasing energy for cellular activities. ***(1 mark)***

ATP is regenerated from ADP by two main processes:

- Oxidative phosphorylation: Occurs in the inner mitochondrial membrane, where ATP is produced by ATP synthase using energy from the electron transport chain. Oxygen acts as the final electron acceptor, making this the major source of ATP. ***(1/2 mark)***
- Substrate-level phosphorylation: Occurs in the cytoplasm (glycolysis) and mitochondrial matrix (Krebs cycle), where ATP is formed directly by transferring a phosphate group to ADP. **(1/2 mark)**



9. Define oxidation.*(2 marks)*

- It is the process in which an atom, ion, or molecule loses electrons. This loss of electrons increases the oxidation state of the substance. In simple terms, oxidation is the "loss of electrons".

10. Define reaction specificity.

Reaction specificity is the property of an enzyme by which it catalyzes only a particular type of chemical reaction, even if it can act on similar substrates.***(1 mark)***

Eg. It specifically catalyzes the decomposition reaction of hydrogen peroxide: $H_2O_2 \rightarrow H_2O + O_2$ Even though other substrates may be present, catalase only performs this specific reaction***. (1 mark)***

Part C

Short Essay Questions

Answer any 4 questions

Each question carries 5 marks

11. Describe transcytosis.*(Any 5 points 1 mark each)*

- Transcytosis is the transport of macromolecules across the interior of a cell.
- It involves uptake of substances into vesicles from one side of the cell (endocytosis).
- The vesicles are transported across the cell.
- Substances are released on the opposite side by exocytosis.
- It is important in cells that form barriers.
- Allows selective transfer of substances without altering their composition.
- Commonly occurs in blood capillaries.
- Proteins and antibodies are transported across endothelial cells.
- Helps in exchange of materials between bloodstream and tissues.
- Maintains structural integrity of the barrier.

12. Explain the various components of ETC.

1. Complex I (NADH dehydrogenase) ***(1mark)***

- Accepts electrons from NADH
- Transfers them to the next carrier
- Pumps protons (H^+) into the intermembrane space

2. Complex II (Succinate dehydrogenase) ***(1mark)***

- Receives electrons from $FADH_2$
- Passes them into the chain
- Does not pump protons

3. Coenzyme Q (Ubiquinone) ***(1/2 mark)***

- A mobile lipid carrier within the membrane
- Transfers electrons from Complex I and II to Complex III

4. Complex III (Cytochrome bc_1 complex) ***(1/2 mark)***

- Transfers electrons to cytochrome c
- Pumps protons across the membrane

5. Cytochrome c ***(1/2 mark)***

- A mobile protein carrier
- Carries electrons from Complex III to Complex IV

6. Complex IV (Cytochrome c oxidase) ***(1/2 mark)***

- Transfers electrons to oxygen
- Oxygen combines with electrons and H^+ to form water
- Pumps protons

7. ATP Synthase (Complex V) ***(1mark)***

- Uses the proton gradient created by the ETC
- Synthesizes ATP from ADP + Pi

13. Explain the Watson Crick model of RNA. ***(each point carries 1 mark)***

1. Single-stranded molecule

- RNA is usually single-stranded, unlike DNA.

2. Sugar-phosphate backbone

- Made of ribose sugar + phosphate groups.

3. Nitrogen bases

- Contains Adenine (A), Uracil (U), Guanine (G), Cytosine (C)
- Uracil replaces thymine (found in DNA)

4. Base pairing (Watson–Crick type)

- When pairing occurs:
    - A pairs with U
    - G pairs with C

5. Secondary structure formation

- RNA can fold back on itself forming hairpin loops and stems due to internal base pairing.

14. List down the factors affecting diffusion rate. ***(Any 5 points 1 mark each)***

- **Temperature**

  Higher temperature → increases molecular energy and movement → faster diffusion

  Lower temperature → decreases molecular energy → slower diffusion
- **Solubility**

  Nonpolar/lipid-soluble substances diffuse faster than polar substances
- **Concentration Gradient**

  Greater concentration difference → faster diffusion
- **Mass of Molecules**

  Heavier molecules → move slowly → slower diffusion

  Lighter molecules → move faster → faster diffusion
- **Density of Solvent**

  Higher density → slower diffusion (more resistance to movement)

  Lower density → faster diffusion
- **Surface Area / Membrane Thickness**

  Increased surface area → faster diffusion

  Thicker membrane → slower diffusion
- **Distance Travelled**

  Greater distance → slower diffusion

  Shorter distance → faster diffusion

15. Explain the stages of oxidation in food. ***(Each stage 1.5 marks each and example 0.5 marks)***

- **First Stage (Primary Metabolism)**

  Digestion in the gastrointestinal tract breaks macromolecules into small units

  Example: Proteins → Amino acids
- **Second Stage (Secondary / Intermediary Metabolism)**

  Digested products are absorbed

  Catabolised to smaller components and oxidised to CO_2

Occurs mainly in mitochondria via the citric acid cycle

Generates reducing equivalents: NADH and $FADH_2$

- **Third Stage (Tertiary Metabolism / Cellular Respiration)**

  NADH and $FADH_2$ enter the electron transport chain (ETC)

  Energy is released during this process

  Energy is used for synthetic activities in the body

16. Explain gene mapping and its classification.

Gene mapping is the process of determining the position (locus) of genes on a chromosome and the distance between them***(1 mark)***

1. Genetic (Linkage) Mapping ***(2 marks)***

- Based on recombination frequency during meiosis
- Distance is measured in centimorgans (cM)
- The more recombination, the farther apart the genes
- Does not give exact physical distance, only relative positions

2. Physical Mapping ***(2 marks)***

- Determines the actual physical location of genes on DNA
- Distance is measured in base pairs (bp)
- More accurate than genetic mapping
- Includes techniques like restriction mapping and sequencing

Part D

Essay Questions

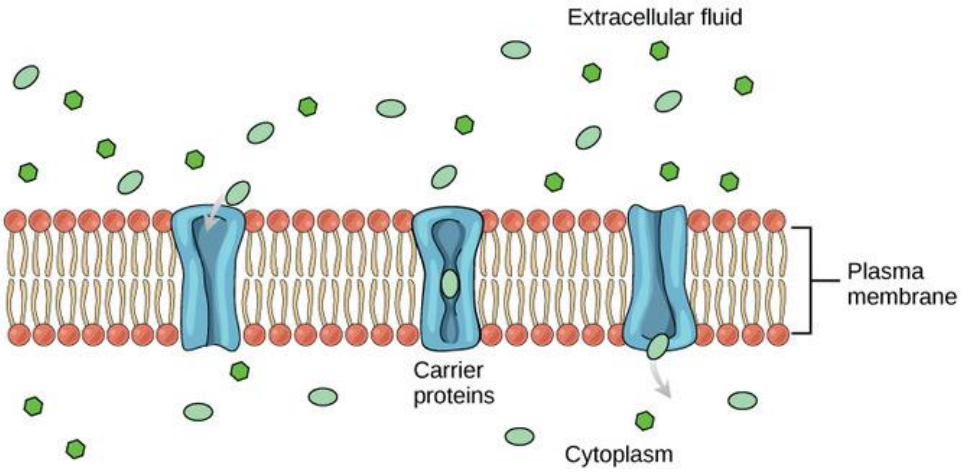
Answer any 2 questions

Each question carries 10 marks

17. Explain facilitated diffusion. ***(Any eight points 1 mark each and diagram 2 marks)***

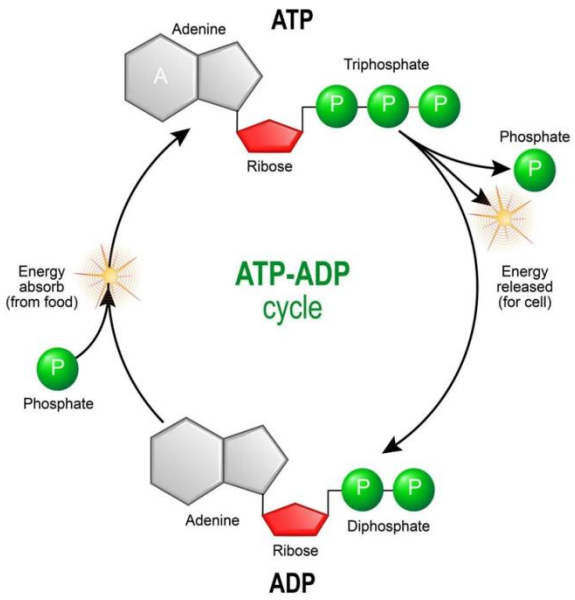
- Facilitated diffusion is a type of passive transport (also called facilitated transport).

- Substances move across the plasma membrane with the help of membrane proteins (transmembrane integral proteins).
- Movement occurs along the concentration gradient (high → low concentration).
- No cellular energy (ATP) is required.
- Mainly involves ions and polar molecules that cannot pass through the hydrophobic lipid bilayer.
- Membrane proteins protect these substances from repulsion by the membrane.
- Substances first bind to specific protein or glycoprotein receptors on the outer surface.
- Then they are transferred to integral proteins for transport across the membrane.
- **Types of transport proteins involved:**
  Channel proteins: form pores (often made of β-pleated sheets) for passage
  Carrier proteins: bind to substances and assist their movement across the membrane



18. Describe the methods of ATP synthesis and ATP-ADP cycle. .

ATP–ADP Cycle ***(2 marks)***



When energy is required, ATP is hydrolyzed:

ATP→ADP+Pi+EnergyATP \rightarrow ADP + P_i + EnergyATP→ADP+Pi+Energy

- The terminal phosphate bond is broken, releasing energy.
- The energy released is used for muscle contraction, active transport, biosynthesis, and other cellular activities.
- ADP is then rephosphorylated to ATP using energy derived from the breakdown of nutrients such as glucose.
- This cycle occurs continuously, ensuring that cells have a constant and immediate supply of energy.

Methods of ATP Synthesis

1. Substrate-Level Phosphorylation ***(4 marks)***

Definition:
Direct formation of ATP by transfer of a phosphate group from a high-energy substrate to ADP.

Sites:

- Cytoplasm → during glycolysis
- Mitochondrial matrix → during Krebs cycle

Examples:

- 1,3-bisphosphoglycerate → 3-phosphoglycerate
- Phosphoenolpyruvate → Pyruvate

Features:

- Does not require oxygen, hence occurs in both aerobic and anaerobic conditions
- Produces a small amount of ATP
- Provides rapid and immediate energy to the cell
- Important in tissues with low oxygen supply and in cells lacking mitochondria

2. Oxidative Phosphorylation ***(4 marks)***

Definition:
ATP synthesis using energy released from electron transfer in the electron transport chain (ETC).

Site:

- Inner mitochondrial membrane

Process:

- Electrons from NADH and $FADH_2$ pass through a series of carriers in the ETC
- Energy released during electron transfer is used to pump protons (H^+) into the intermembrane space
- This creates an electrochemical proton gradient (proton motive force)
- Protons flow back into the matrix through ATP synthase, driving the formation of ATP from ADP and Pi
- This mechanism is explained by the chemiosmotic theory

Role of Oxygen:

- Oxygen acts as the final electron acceptor
- It combines with electrons and protons to form water
- This prevents the accumulation of electrons and allows the chain to continue functioning

Features:

- Produces the maximum amount of ATP (~28–32 ATP per glucose)
- Requires oxygen, hence it is an aerobic process
- Highly efficient and supplies most of the energy required for cellular activities
- Occurs only in cells with functional mitochondria

19. Explain the major types of DNA damages and repair.

Types of DNA Damage ***(5 marks)***

1. Point mutations
A change occurs in a single base of DNA, which may involve substitution, insertion, or deletion, potentially altering gene function.

2. UV-induced damage
Ultraviolet radiation causes the formation of thymine dimers, which distort the DNA helix and interfere with replication.

3. Chemical damage
Chemicals can modify DNA bases through reactions such as deamination and alkylation, leading to improper base pairing.

4. Oxidative damage
Free radicals generated during metabolism damage DNA bases and the sugar-phosphate backbone.

5. Double-strand breaks
Both strands of the DNA are broken, making it one of the most severe and potentially lethal forms of damage.

DNA Repair Mechanisms ***(5 marks)***

1. Direct Repair
This mechanism directly reverses DNA damage without removing the base, such as the photoreactivation of thymine dimers.

2. Base Excision Repair (BER)
Damaged bases are recognized, removed, and replaced with the correct bases by specific enzymes.

3. Nucleotide Excision Repair (NER)
A segment of damaged DNA is removed and resynthesized, especially in cases like thymine dimer formation.

4. Mismatch Repair (MMR)
Errors introduced during DNA replication are identified and corrected to maintain accuracy.

5. Double-Strand Break Repair
DNA breaks are repaired either by homologous recombination, which is accurate, or by non-homologous end joining, which is less precise.

6. SOS Repair (Error-prone repair)
Under severe DNA damage, this system allows replication to continue using low-fidelity polymerases, though it may introduce mutations.

20. Explain the process of protein biosynthesis in humans

Protein Biosynthesis in Humans ***(1 mark)***

Protein biosynthesis occurs in two major steps: transcription and translation, followed by post-transcriptional and post-translational modifications.

1. Transcription (in nucleus) ***(3 marks)***

Definition:
Synthesis of RNA from DNA using RNA polymerase.

Steps of Transcription:

a. Initiation

- RNA polymerase binds to the promoter region of DNA
- DNA unwinds to expose the template strand

b. Elongation

- RNA polymerase moves along the DNA template (3′ → 5′)
- Synthesizes pre-mRNA in 5′ → 3′ direction
- Complementary base pairing occurs (A–U, G–C)

c. Termination

- RNA polymerase reaches the termination signal
- Pre-mRNA is released

Post-Transcriptional Modifications (in nucleus) ***(2 marks)***

The primary transcript (pre-mRNA / hnRNA) is processed into mature mRNA:

1. 5′ Capping

- Addition of 7-methyl guanosine cap
- Protects mRNA and helps ribosome binding

2. 3′ Poly-A Tail

- Addition of adenine nucleotides
- Increases stability of mRNA

3. Splicing

- Removal of introns (non-coding regions)
- Joining of exons (coding regions)

2. Translation (in cytoplasm) ***(3 marks)***

Definition:
Synthesis of protein from mRNA on ribosomes.

Steps of Translation:

a. Initiation

- Small ribosomal subunit binds to mRNA
- Recognizes start codon (AUG)
- Initiator tRNA carrying methionine binds
- Large subunit joins to form complete ribosome

b. Elongation

- tRNA brings specific amino acids to ribosome
- Codon–anticodon pairing ensures correct sequence
- Peptide bonds form between amino acids
- Ribosome moves along mRNA (translocation)

c. Termination

- Ribosome reaches a stop codon (UAA, UAG, UGA)
- Release factors bind
- Polypeptide chain is released

Post-Translational Modifications ***(1 mark)***

After translation, the protein becomes functional through modifications:

1. Folding

- Protein folds into its 3D structure (often with chaperones)

2. Chemical modifications

- Phosphorylation, glycosylation, acetylation, etc.

3. Cleavage

- Some proteins are cut into active forms

4. Transport

- Proteins are sent to specific locations (ER, Golgi, etc.)